Course: Software Quality Engineering (Lab)

Lab 01: GitHub Repository and SQE Project Setup

Objective: Learn how software-quality activities are integrated into a professional GitHub-based development workflow.

Tasks:

1. **Create a GitHub repository named:**
   ‘sqe-library-management’

2. **Initialize the repository with:**
   README.md
   .gitignore
   LICENSE
3. **Create the following directory structure:**
```
sqe-library-management/
|
├── src/
├── tests/
├── docs/
├── screenshots/
├── .github/
|   └── workflows/
|
├── README.md
├── LICENSE
└── .gitignore
```

4. **Add a project description to `README.md`.**
5. **Create at least three GitHub Issues:**
   Implement Book Management
   Implement Member Management
   Develop Book Search

6. **Create GitHub labels:**
   bug
   enhancement
   testing
   documentation
   high-priority
   quality

7. **Create a GitHub Project and add the issues**.

Deliverables at the end of Lab 01 completion:

- GitHub repository URL
- Repository structure
- Three or more issues
- GitHub Project board
- README file

What is Git?
What is GitHub?
What is the difference between Git and GitHub?
Why should .gitignore be used?
What is an issue in GitHub?
Why are labels useful for SQE?

Steps for Lab 01:

Task 1 — Create a GitHub Account

Task 2 — Install Git: Download and install Git from the official Git Website

After installation:

Git Bash (Windows)

git --version

Task 3 — Configure Git:

git config --global user.name "Your Name"

git config --global user.email your-email@example.com

Verify the configuration: git config --global –list

Note: Use the email associated with your GitHub account, or configure GitHub's privacy-preserving email if you prefer not to expose your personal email in commits.

Task 4: Create the GitHub Repository

‘sqe-library-management’

Repository Settings

Select:

- **Public or private according to your instructor's requirements**
- **Add README**
- **Add .gitignore**
- **Select the appropriate programming language if applicable**
- **Add a license if required**

Click: Create repository

Task 5 — Create Project Structure

```
sqe-library-management/
|
├── src/
├── tests/
├── docs/
├── screenshots/
|
├── README.md
├── .gitignore
└── LICENSE
```

Directory	Purpose
src	Application source code
tests	Test cases and automated tests
docs	Project documentation
screenshots	Evidence of testing/activities
README.md	Project information
.gitignore	Files Git should not track
LICENSE	Project licensing information

Task 6 — Clone the Repository

Copy the repository's HTTPS URL from GitHub.

git clone <repository-url>

cd sqe-library-management

Task 7 — Create the First Project Documentation

Open `README.md`. and Write Project Description

Task 8 — Check Changes

git status: Git should report that `README.md` has been modified.

git dif: shows the difference between the previous version and the current working version.

Task 9 — Stage the Change

git add README.md

git status

Task 10 — Commit the Change

git commit -m "docs: add project README"

The change is now recorded in the local Git repository.

git log –oneline: You should see your commit.

Task 11 — Push to GitHub

git push

Refresh the GitHub repository in your browser and view the updated README should now be visible.